

Digital LIMS: Automated Quality Control & SPC Analytics Pipeline

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Target Application: Fertilizer Companies / Industrial Chemical Manufacturing MTO & Lab Analyst Roles

1. Executive Summary & Business Value

In large-scale continuous chemical synthesis plants (such as Urea and Ammonia production complexes operated by FFC, Engro, or Fatima), the Quality Control (QC) laboratory serves as the central operational checkpoint. Chemical analysts execute regular shift-wise testing for critical quality parameters including Urea Biuret Content (%), Moisture Content (%), Prill Size Distribution, Ammonia Stream Purity (%), and High-Pressure Boiler Feed Water pH.

Operational Pain Point & Solution: Traditional manual data entry into basic spreadsheets leads to delayed anomaly detection, human error during routine statistical calculations, and high re-processing overheads when off-spec product reaches main storage silos. This project bridges wet-lab chemical analytical workflows with Python-driven Statistical Process Control (SPC) and an interactive executive Power BI dashboard.

2. System Architecture & Workflow Pipeline

The Digital LIMS pipeline operates as a seamlessly integrated 4-tier data processing engine:

- Tier 1 - Industrial Lab Ingestion:** Simulates 1 year of continuous 3-shift operation (1,095 total testing records) capturing 5 critical chemical parameters and analyst IDs.
- Tier 2 - Python Analytical Engine (Jupyter Notebook):** Executes automated data cleaning, multi-parameter spec checks, 3-Sigma Shewhart Control Limit calculations, and Process Capability Index (Cpk) modeling.
- Tier 3 - Structured Data Export:** Generates enriched CSV datasets containing calculated batch status flags (PASS/FAIL) and individual parameter violation tags.
- Tier 4 - Power BI Executive Dashboard:** Provides real-time visualization of shift-wise compliance, overall pass rates, parameter failure distributions, and interactive analyst auditing.

3. Technology Stack & Operational Rationale

Tool / Technology	Primary Purpose	Why It Was Chosen / Industrial Value
Python (Pandas, NumPy)	Automated Data Pipeline	Replaces manual entry. Programmatically enforces multi-parameter thresholds across large historical datasets without human error.

SciPy & Matplotlib	Statistical Quality Control (SQC)	Enables exact calculation of Cpk process capabilities and generates publication-grade 3-Sigma X-bar Shewhart control charts.
Power BI Desktop	Executive Visual Management	Translates complex chemical data into an interactive control-room dashboard with dynamic slicers for Plant Managers and QC Heads.
DAX Measures	Dynamic KPI Aggregation	Ensures mathematically precise KPI calculations (e.g., distinguishing simple parameter averages from weighted batch compliance rates).

4. Analytical Chemistry & Quality Assurance Methodology

The pipeline evaluates five critical laboratory indicators mapped directly to standard industrial fertilizer manufacturing SOPs:

- 1. Urea Biuret Content (< 1.00% USL):** Biuret (C2H5N3O2) is an undesirable toxic byproduct formed via the endothermic dimerization of urea at elevated temperatures in the synthesis reactor loop. High biuret severely harms crop foliage.
- 2. Moisture Content (< 0.30% USL):** Excess moisture in final urea prills causes lumping, mechanical caking in storage silos, and severe degradation during transfer and bagging.
- 3. Prill Size Distribution (> 90.0% in 1.0mm - 2.8mm):** Crucial for physical application in agricultural broad-casters. Non-uniform prill size leads to uneven soil nutrient delivery.
- 4. Ammonia Stream Purity (> 99.5% LSL):** Supplied from the ammonia synthesis/recovery block. Low purity indicates unreacted carbamate buildup or inert gas accumulation in the loop.
- 5. Boiler Feed Water pH (8.5 - 9.5 Target Window):** Strictly monitored in high-pressure steam boilers to prevent acidic line corrosion (< 8.5) or caustic scaling (> 9.5).

Statistical Process Control (SPC) Formulas

- Process Capability Index (Cpk):** Measures process alignment relative to specification limits.

$$Cpk = \min\left(\frac{USL - \mu}{3\sigma}, \frac{\mu - LSL}{3\sigma}\right)$$

- 3-Sigma Upper & Lower Control Limits (UCL / LCL):**

$$UCL = \mu + 3\sigma \quad | \quad LCL = \mu - 3\sigma$$

5. Key Performance Indicators (KPIs) & Audit Results

Analysis of 1,095 historical shift records revealed critical insights into plant operational health:

Metric / Indicator	Audited Value	Operational Significance / Finding
Total Shift Tests Audited	1,095	Complete 365-day dataset covering Morning, Evening, and Night shifts.

